

# JEE (Advanced) 2026 Chemistry Paper-2

## Answer Key with Detailed Solutions

| Question | Answer  |
|----------|---------|
| Q1       | B       |
| Q2       | C       |
| Q3       | B       |
| Q4       | D       |
| Q5       | A, C    |
| Q6       | A, C    |
| Q7       | B, C    |
| Q8       | A, B, D |
| Q9       | B, C, D |
| Q10      | 2       |
| Q11      | 0.03    |
| Q12      | 50      |
| Q13      | 8       |
| Q14      | 86000   |
| Q15      | 1246    |
| Q16      | 0.05    |
| Q17      | 40      |
| Q18      | 4.66    |

### Q1

Using limiting molar conductivity and conductivity relation:  $\Lambda_m = \kappa \times 1000 / C$ . Substituting the conductivity of saturated AgCl solution gives solubility  $X \approx 10^{-4} \text{ mol L}^{-1}$ . Therefore  $\log_{10}(X^{-1})=4$ .

### Q2

ONO bond angle increases with decreasing lone pair repulsion. Correct order:  $\text{NO}_2^- < \text{NO}_3^- < \text{NO}_2 < \text{NO}_2^+$ .

### Q3

Natural rubber on ozonolysis gives methyl glyoxal, which gives positive Tollens' and iodoform tests. On heating with NaOH it undergoes Cannizzaro-type reaction giving the corresponding hydroxy acid product.

### Q4

The artificial sweetener described is sucralose. Correct structure corresponds to option D.

### Q5

For first-order reaction,  $\ln[R]$  vs  $t$  is linear and  $[P]$  increases exponentially toward limiting value. Hence A and C are correct.

**Q6**

P contains two lone pairs on central atom. Q behaves as fluorinating agent. Hence A and C are correct.

**Q7**

Ionic radius order is  $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Na}^+$ . Density of potassium is greater than sodium because of packing and atomic mass considerations. Hence B and C are correct.

**Q8**

Q gives aromatic aldehyde with ethanol. S gives phthalein dye test and T is colored due to conjugation. Hence A, B and D are correct.

**Q9**

Fructose gives Fehling's test due to enediol rearrangement. Invert sugar is equimolar glucose and fructose with net rotation about  $-40^\circ$ . Hence B, C and D are correct.

**Q10**

Using hydrogen-like transition energy relation and wavelength ratios gives the lowest possible value of  $a+b = 2$ .

**Q11**

Using  $\text{pH} = 3$  and  $K_a = 10^{-5}$ , equilibrium concentration is calculated. Applying Freundlich isotherm with slope 1 gives  $k = 0.03 \text{ L mol}^{-1}$ .

**Q12**

Using boiling point elevation:  $\Delta T_b = iK_b m$ . Degree of dissociation obtained from van't Hoff factor gives 50% dissociation.

**Q13**

Counting triangular faces containing one N atom and two Cl atoms in both octahedral complexes gives total = 8.

**Q14**

Polymer formed is biodegradable polyester. Using stoichiometry of 500 mol each monomer and repeating unit molar mass gives total polymer mass = 86000 g.

**Q15**

Using ideal gas molar volume at 300 K and liquid molar volume from density, ratio  $\approx 1246$ .

**Q16**

Using Raoult's law and total vapour pressure relation, mole fraction of B in vapour phase = 0.05.

**Q17**

Kjeldahl nitrogen estimation gives  $\text{NH}_3$  evolved corresponding to 40 mL of 1 M  $\text{H}_2\text{SO}_4$  required for neutralization.

**Q18**

Using sulphur estimation by Carius method and stoichiometry of  $\text{BaSO}_4$  formation gives 4.66 g of  $\text{BaSO}_4$ .